

Co1 - AT 3 ASSIGNMENT

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Course : Ethical Hacking

Course I TA 1435

Code :

Assignment

1) Ethical Resolution of Vulnerability Exploitation Conflict:

Introduction:-

In this situation, I am performing an authorized penetration test on a hospital's critical patient-monitoring system. I discover a serious vulnerability that could cause the system to crash if exploited. Although the contract requires "full exploitation to confirm impact", exploiting it could put patients at risk.

Ethical Issues:-

- 1) Patient Safety: A hospital's monitoring system may be directly related to patient care. A crash could cause serious harm.
- 2) Contractual obligation: The penetration tester has agreed to fully exploit vulnerabilities as part of the contract.
- 3) Professional Responsibility: Security professionals have a duty to avoid unnecessary damage while testing systems.
- 4) Risk Versus Benefit: Completely exploiting the vulnerability may provide stronger evidence, but the potential harm is much greater.
- 5) Confidentiality and integrity: Testing must not expose or modify sensitive patient information unnecessarily.

Ethical Course of Action:

I would not perform the full exploitation on the live patient-monitoring system.

→ First, Stop exploitation before reaching a dangerous stage.

→ Immediately inform the client and the hospital's security IT team about the vulnerability and its potential impact.

→ Collect only the minimum evidence required to prove that the vulnerability exists.

→ If possible, reproduce the vulnerability in a test, staging, or isolated environment.

Balancing Contract and Ethics:-

A contract is important, but it does not justify an action that could unnecessarily endanger human lives.

The tester should interpret the requirement for full exploitation in the context of safe and responsible security testing.

The tester should communicate the risk and seek an alternative method of confirming the vulnerability.

Conclusion :-

The correct ethical decision is to prioritize patient safety while still providing sufficient evidence of the vulnerability. Full exploitation should be avoided on the live system if it could cause harm.

Testing is a controlled environment, minimum-impact verification, proper documentation and immediate communication with client provide a balance.